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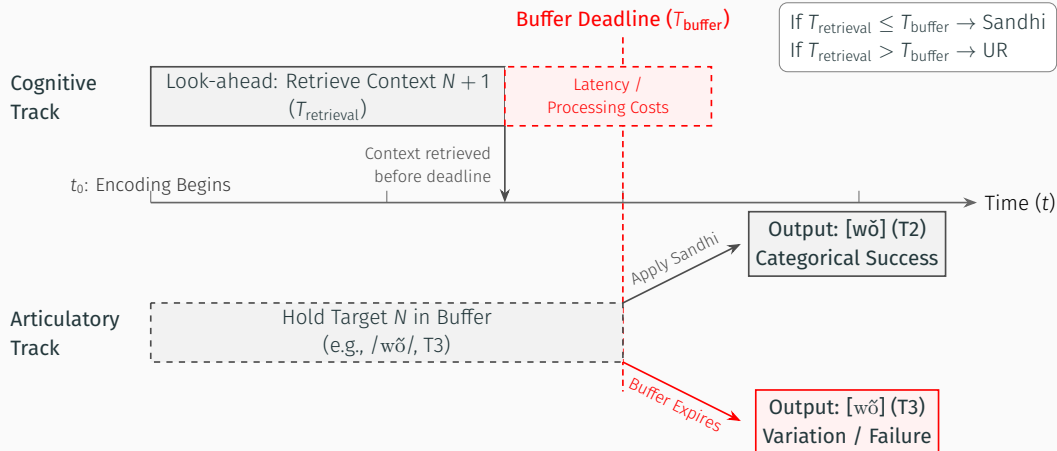
- Phonetics and Phonology
- Phonology and psycholinguistics
- Speech planning and production

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How much should we encode gradience and variation into phonological grammar?

- Probabilistic models of phonology
- Importing non-categorical representations into Phonology
- **Processing**, e.g., frequency and retrieval, memory and decay, activation and competition, etc.
 - English and English-Spanish tapping (Kilbourn-Ceron 2016, Hironori et al. 2025)
 - French Liaison (Kilbourn-Ceron 2017)
 - Coronal stop deletion (Tanner et al. 2017)
 - Aux contraction (MacKenzie 2012, 2016)

Anticipatory processes: Overflow of the articulatory buffer

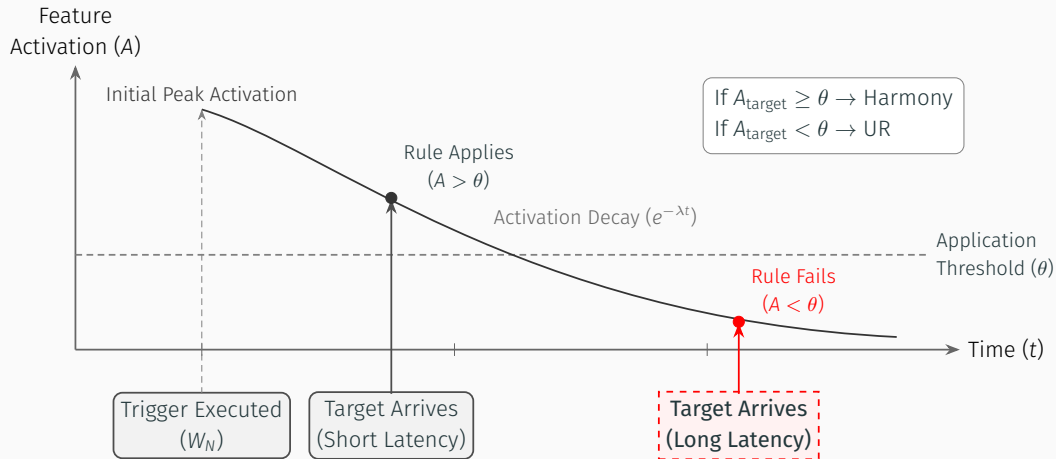


Hypothesis: Phonological variation is the outcome of a temporal race between active lexical look-ahead and a capacity-limited articulatory buffer. When the cognitive cost of structural look-ahead exceeds this temporal limit, categorical rules fail.

Current projects:

- Pre-articulatory planning window in sandhi processes
(Preliminary MA Thesis topic)
- Output oriented articulatory compensation with competing gestures
(Ongoing, with Yao Yao and Tianqi Ju, PolyUHK)

Perseverative processes: Activation decay



Goal: Ultimately, by shifting the burden of variation to temporal limits (i.e., the race and the decay), we can maintain a clean, categorical grammar while also explaining gradience and variation.

By integrating these psycholinguistic constraints with phonetic realities (e.g., cue weighting/trading, phonetic knowledge, and post-stress superiority, etc.), we can develop a more principled account of the architecture of phonological grammar.